

Week 1: High-Level Architecture Design

PW16_MG_01

Purpose and Context

The first week of the capstone project focused on establishing a high-level architectural direction before engaging in detailed design or implementation. This phase marked the conceptual initiation of the project and was intended to ensure that all subsequent work would be guided by a coherent and defensible system structure. The emphasis remained on defining *how the system should be organized*, rather than *how individual components should be built*.

Architectural Thinking and Key Decisions

The system was conceptualized as a layered architecture with well-defined responsibility boundaries across data handling, intelligence, explainability and governance, and user interaction. This approach supports modularity and long-term maintainability.

Design Philosophy and Rationale

We split up decision-making and explaining decisions in our system's design. This makes the system easier to understand and check later, without messing with the core reasoning. The setup focuses on keeping different parts separate, so changes in one area don't mess with others.

We're going for clarity and making sure things are easy to follow and can be changed later, instead of just rushing to get it done quickly.

Assumptions, Constraints, and Risks

Early assumptions included the need for scalability and independent evolution of system components. Academic timelines and evaluation constraints were acknowledged, reinforcing the importance of minimizing architectural rework in later phases. Addressing these considerations early reduces downstream risk.

Key Architectural Decisions:

- We went with a layered architecture, keeping responsibilities clear.
- We made explainability a main part of the architecture.

Next Steps:

- Decompose each architectural layer into functional components.
- Decide on mechanisms to develop actual explanation and rule-keeping systems.

Continuity and Forward Alignment

This architecture is now the foundation for the next phase, where we'll break things down further. This will keep the project consistent and on track.

None

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SRN:

plan: For next week -

Updates

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Week 2 Capstone Progress Report

PW26_MG_02

Hybrid HFT System Design and Explainability Roadmap

During Week 2 of the capstone project, substantial conceptual and architectural progress was made toward designing a production-grade hybrid High-Frequency Trading (HFT) system. The proposed system uniquely blends the low-latency execution advantages of C++ with the analytical flexibility and rapid experimentation capabilities of Python. The core architecture is structured around a six-layer quantitative model pipeline, encompassing statistical models, machine learning techniques, deep learning frameworks, reinforcement learning strategies, execution models, and market microstructure analysis. In addition, the system explicitly integrates ten critical trading enhancements, including signal normalization, transaction cost-aware filtering, drawdown-based kill switches, and realistic backtesting constraints. These design decisions are aimed at ensuring robustness, risk awareness, and real-world deployability. In the upcoming weeks, this trading engine will be evolved into the foundation of an explainability framework, enabling in-depth analysis and interpretation of strategy behavior, particularly under highly volatile market conditions, which is a central objective of the capstone project.

Polyglot Architecture Exploration and Integration

Following the core system design, a focused effort was made to explore the feasibility of a polyglot architecture supporting multiple programming ecosystems, including Python, C++, Java, R, and the MERN stack. A small-scale prototype was developed to validate interoperability among these heterogeneous components. Each language-specific module was designed to produce structured outputs, which were then standardized and merged using JSON as a common interchange format. This approach facilitated seamless communication across modules while retaining the individual strengths of each language, such as Python's modeling flexibility, C++'s performance efficiency, Java's stability, R's statistical capabilities, and MERN's suitability for web-based interfaces. Although exploratory, this work established a strong foundation for a modular, extensible system architecture capable of supporting complex quantitative workflows.

Live Data Collection and Architecture Development

In parallel with architectural experimentation, a live data collection process was actively designed and implemented during this week. The objective was to establish a reliable, production-ready pipeline capable of continuous data ingestion for real-time analysis and validation. Live market data collection commenced on 14th January and has been consistently operational through 17th January. The pipeline emphasizes data accuracy, precise timestamping, and structured storage to ensure compatibility with downstream modeling, backtesting, and execution components. **In subsequent weeks of the capstone project, this pipeline will remain active with a primary focus on continuous data collection, ensuring the availability of consistent, real-world datasets for analysis and validation, without introducing additional architectural changes during this phase.** This live data architecture is a critical component of the capstone, as it enables realistic evaluation of system performance under real market conditions rather than relying exclusively on historical or simulated datasets.


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Week 3 Progress Documentation

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Building on the feasibility analysis and system-level planning completed in Week 1 and Week 2, Week 3 marked the transition from abstraction to visibility. With the problem formulation, high-level architecture, and implementation roadmap already established, the focus shifted to making the system tangible—something that could be seen, interacted with, and reasoned about. This phase emphasized understanding how the system should behave in practice, rather than questioning its viability.

Why We Began with a Frontend Representation

To enable this transition, we began developing the frontend as an early, exploratory representation of the system. This was a deliberate design decision. Instead of treating the interface as an end-stage deliverable, we used it as a working canvas to visualize workflows, interactions, and decision boundaries. By externalizing these elements, the implementation roadmap could be actively tested against realistic usage scenarios, strengthening confidence in the overall design.

In this sense, the frontend functions as a discovery tool—one that sharpens backend requirements rather than constraining them prematurely.

Structuring the System Through Interface Design

The interface was structured to clearly delineate market visualization, execution flow, portfolio intelligence, and system trust components. Each section represents a distinct functional responsibility, making architectural intent immediately visible and reproducible. This modular structure also supports scalability and ease of future extension.

Market Views, Indicators, and Execution Flow

Time-based market views and technical indicators were implemented with explicit behavior and clear labeling. Indicators were kept independent to preserve interpretability, revealing data dependencies, update frequencies, and historical depth requirements. The execution panel was designed as a staged, intentional process, culminating in a swipe-based confirmation that encodes deliberate action and informs backend validation and risk-check logic.

Refinement as a Path to Reproducibility

As the system becomes more visible, refinement accelerates. We continue to fine-tune the frontend to deepen clarity around backend logic, data contracts, and edge cases. At this stage, the interface serves as a living specification—one that supports reproducibility and clearly maps the system's forward trajectory.

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Week 4 Progress Documentation

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Building on the architectural foundation established in Week 1, the hybrid system design and feasibility exploration carried out in Week 2, and the interface-driven system visibility achieved in Week 3, Week 4 focused on consolidating the project into a coherent and communicable system representation. The intent of this phase was not to introduce new components, but to ensure that everything developed so far can be clearly reasoned about, explained, and defended as a unified whole. This step was essential in transforming individual design decisions into a system that reflects deliberate control and intellectual ownership.

Articulating Intent and Control

During this week, we clearly defined the abstract and scope, positioning the project as an effort to build a trustworthy, adaptive, and explainable stock prediction platform—one that values accountability, transparency, and restraint as much as raw performance. We explicitly documented Phase-1 feedback and our responses to it, including improved architectural clarity, stronger module separation, UML-based workflow representations, and frontend prototypes that demonstrate system feasibility and real-world alignment.

Our literature survey anchors the project in recent research on adaptive reinforcement learning, causal modeling, risk-aware portfolio management, and explainable financial AI. This grounding reinforces why an explainability-first direction is not merely a design preference, but a necessary response to the limitations of opaque, accuracy-driven financial systems operating under volatile market conditions.

An Inspiring View of the Proposed Methodology

At the heart of our system lies a simple but powerful belief: a financial system should know when to act—and when not to. Our proposed methodology embodies this through an explainability-first controlled execution model, where decisions are taken only when confidence thresholds and market regime stability are explicitly verified. Every execution is paired with a clear rationale, making system behavior auditable, interpretable, and trustworthy.

Equally important is our abstention-first safeguard mode. Instead of forcing predictions under uncertainty, the system chooses restraint, prioritizing capital protection and transparency over aggressive automation. By treating abstention as a deliberate, explainable decision rather than a failure, we design for responsibility, not just intelligence. Together, these modes reflect how we believe real financial systems should behave—measured, self-aware, and accountable.

Outcome

By the end of Week 4, we had not only documented our work, but strengthened our conviction in it. This clarity positions us to treat the ISA-1 presentation in Week 5 as a checkpoint rather than a finish line, where feedback and scrutiny will directly inform sharper design decisions, tighter assumptions, and deeper system refinement in the weeks ahead.

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Week 5 Capstone Progress Documentation

Project ID : PW26_MG_02

Overview

During Week 5, we presented our Phase 2 – Project Progress Review #1 (Review-1) for our capstone project titled "Toward Trustworthy Stock Prediction: A Self-Aware Multimodal Quant AI Framework." This review marked a key transition from conceptual groundwork to structured execution. The presentation covered the project motivation, scope, design philosophy, system architecture, and the planned roadmap for Phase 2.

Problem Motivation and Scope

We articulated the **problem statement** by highlighting the limitations of traditional quantitative and modern AI-based financial prediction systems, including rigidity, lack of transparency, limited accountability, and poor adaptability under volatile market conditions. Our project aims to address these challenges by building a **trustworthy, explainable, and self-aware AI-driven prediction framework** that emphasizes transparency and confidence-aware decision-making.

The **abstract and scope** highlighted the growing need for adaptive and auditable AI systems in regulated financial environments. We demonstrated how explainability, accountability, and adaptability together address investor distrust and regulatory expectations often unmet by opaque models.

Literature Insights and Research Gaps

Through the **literature survey**, we established that existing models perform adequately under stable conditions but struggle in dynamic regimes due to limited explainability and adaptability. This directly motivated our explainability-first architectural design. We identified key research gaps, including data provenance issues, lack of confidence estimation and contextual reasoning, absence of real-time adaptability, and weak accountability and auditability mechanisms.

Design Approach and Methodology

We presented a **frontend-first, modular, and polyglot design approach**, supported by UML diagrams and workflow visualizations. The core philosophy is to execute decisions only under high confidence with transparent justification and abstain under uncertainty to preserve system integrity, treating explainability as the execution gate.

Architecture and Phase 2 Planning

A layered system architecture was introduced, along with implementation challenges and resource requirements, particularly GPU acceleration. A structured Phase 2 progress plan was also outlined.

Panel Feedback and Reflection

The panelists appreciated the architectural clarity and explainability-driven design, recommending the development of an **explainability toolkit** in upcoming phases. They noted the project's potential to revolutionize the future of finance.

Overall, Week 5 was an enriching experience that strengthened our direction for Phase 2.

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Week 6 Capstone Progress Documentation

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Overview

During Week 6, we strengthened our project validation by incorporating real-world industry insights, as recommended by the panel during Phase 2 Review-1. To achieve this, we interacted with our college alumni who has worked on a similar GenAI-driven investment reasoning platform called Kshana. This discussion was highly valuable, as it confirmed that our capstone project is addressing a critical and still unsolved problem in trustworthy financial AI.

Industry Background: Kshana

Our alumni introduced us to Kshana (Sanskrit: क्षण), meaning a "moment in time" or an instant of heightened awareness. Kshana positions itself as a next-generation financial intelligence platform that goes beyond answering "what is happening in markets" (prices, fundamentals, flows) and instead focuses on uncovering "why it is happening."

Powered by its proprietary Foundational Investment Reasoning Engine (FIRE), Kshana's AI agents continuously monitor market conditions, extract economically meaningful signals, and propose next-best actions and trade setups in real time. The alumni also shared how the company is backed by experienced founders and prominent advisors from global finance and technology ecosystems.

Key Discussion: Validation of Our Project Problem

During our conversation, the alumni appreciated the direction of our project and agreed that we are solving a genuine limitation that even advanced industry products like Kshana still face today. They emphasized that while GenAI investing systems are promising, several core challenges remain unresolved.

Major Limitations Highlighted

1. Model Risk and Hallucination

The alumni explained that LLM-based reasoning can hallucinate causal links, overfit narratives, or confuse correlation with causation. In finance, a convincing but wrong explanation can be more harmful than no explanation.

2. Regulatory-Grade Explainability Gap

They also pointed out that regulators demand deterministic and audit-ready decision paths, which current reasoning engines cannot always guarantee.

3. Dependency on Data Quality

Since these systems ingest broker reports, earnings transcripts, and news, biased or misleading inputs can significantly impact outputs, reinforcing the garbage-in-garbage-out issue.

4. Structural Market Challenges

The alumni discussed alpha decay, trust barriers among senior fund managers, latency constraints for HFT use-cases, and competitive pressure from incumbents like Bloomberg.

Conclusion and Reflection

Overall, Week 6 was a major milestone in strengthening the industry relevance of our capstone project. Our alumni interaction confirmed that the limitations we aim to solve—hallucination control, regulatory explainability, and trustworthy decision-making—remain urgent and unsolved even in cutting-edge GenAI investing platforms. This strongly validates the long-term impact potential of our work.

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Week 7 Capstone Progress Documentation

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Week 7 marks one of the most defining and challenging phases of our application's journey. We have identified a **critical unsafe JSON deserialization vulnerability** deeply embedded within our request processing and data transformation layers. This discovery represents not just a technical issue, but a pivotal moment — a turning point where we choose resilience over risk and long-term integrity over short-term convenience.

This vulnerability exposes the system to severe threats, including remote code execution, unauthorized access, and potential full service compromise. Rather than applying temporary patches, we are committing to a fundamental architectural shift. This is not a minor fix — it is a complete redesign of how we define trust boundaries, validate input, and handle data across services.

Our remediation strategy involves eliminating unsafe parsing mechanisms, enforcing strict schema validation, redesigning deserialization workflows, and implementing controlled object mapping. Comprehensive regression testing, security validation, and penetration testing will ensure that every integration point is hardened. This transformation requires careful execution, and therefore remediation will continue until **7th March**, as security remains the highest priority for any application built to endure.

Until full remediation is achieved, we have implemented interim safeguards including enhanced monitoring, stricter request filtering, and reduced exposure of vulnerable endpoints. The system is being actively guarded while we rebuild it stronger.

This phase represents the most challenging shift in our development lifecycle — a conscious move toward security-first engineering. It reinforces a core principle: trust must be earned through design, not assumed through code. By confronting this vulnerability head-on, we are not just fixing a flaw — we are elevating the foundation of our entire system.

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Week 8 – Capstone Progress Documentation

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Week 8 became a defining phase in our journey. The unsafe JSON deserialization vulnerability we identified was not merely a technical defect — it revealed a structural weakness with the potential for remote code execution, aligning with critical insecure deserialization risks recognized by OWASP. This discovery compelled us to confront a fundamental principle: a system's strength depends entirely on the trust boundaries it deliberately enforces.

Balancing Academics and Architecture

As ISA examinations approached, the pressure intensified. However, rather than allowing academic demands to slow remediation, we adopted structure over stress. Mornings were reserved for focused revision and conceptual reinforcement. Evenings became disciplined security refinement sessions. Each day was segmented with precision, ensuring consistent academic preparation while advancing architectural correction.

This balance was not incidental — it was intentional. It reflected our commitment to excellence in both domains.

Technical Reinforcement and Controlled Refactoring

From a technical perspective, Week 8 focused on the controlled isolation of vulnerable deserialization paths. Dynamic object binding was replaced with strict schema-driven DTO validation. Reflection-based instantiation patterns were reviewed and eliminated where necessary. Logging mechanisms were strengthened, and trust boundaries were clearly redefined.

Every modification was executed with care to preserve business logic stability while systematically reducing the attack surface. The objective was not speed, but precision.

This week reinforced an essential insight: innovation is not simply about adding capabilities — it is about fortifying foundations under pressure.

Upcoming Week – Execution with Discipline

In the upcoming week, we will continue balancing ISA examinations with structured security remediation. We will conduct comprehensive regression testing, simulate adversarial payloads aligned with OWASP deserialization risks, reassess trust boundaries, and enhance monitoring mechanisms.

Our goal is clear: to demonstrate disciplined execution, ensuring both academic excellence and architectural resilience without compromise.

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